13. Mystery

> The truth is in the depths.
>
> – Democritus[1]

I've described what I think is the nature of things in the light of what we have learned to date. I've summarized the development of some key ideas of fundamental physics, and I have illustrated the great discoveries made by physics in the twentieth century and the image of the world emerging from the research into a quantum theory of gravity.

Am I sure about all this? I am not.

One of the very first and most beautiful pages in the history of science is the passage in Plato's *Phaedo* in which Socrates explains the shape of the Earth.

Socrates says he 'believes' the Earth is a sphere, with great valleys where men live. He's basically right, if a bit confused. He adds, 'I'm not sure.' This page is worth much more than all of the nonsense on the immortality of the soul which fills the rest of the dialogue. It is not just the oldest text to come down to us which speaks explicitly of the fact that the Earth must be spherical. More importantly, it shines with the crystalline clarity with which Plato acknowledges the *limits* of the knowledge of his time. 'I'm not sure,' says Socrates.

This acute awareness of our ignorance is the heart of scientific thinking. It is thanks to this awareness of the limits of our knowledge that we have learned so much. We are not certain of all which we suspect, just as Socrates was not sure of the spherical nature of the Earth. We are exploring at the borders of our knowledge.

Awareness of the limits of our knowledge is also awareness of the fact that what we know may turn out to be wrong, or inexact. Only by keeping in mind that our beliefs may turn out to be wrong is it possible to free ourselves from wrong ideas, and to learn. To learn something, it is necessary to have the courage to accept that what we think we know, including our most rooted convictions, may be wrong, or at least naïve: shadows on the walls of Plato's cave.

Science is born from this act of humility: not trusting blindly in our past knowledge and our intuition. Not believing what everyone says. Not having faith in the accumulated knowledge of our fathers and grandfathers. We learn